Application of the SPHY Bare-Metal Kernel with 1200 Qubits and Quantum Resilience

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Abstract

The noise applied herein is at level 1.0, demonstrating that the challenge encountered stems not from silicon physics, but rather from the algorithmic model that disregards the core coherence generator factor: Gravity. This paper presents the architecture and experimental results of a zero-heap C99 Bare-Metal microkernel operating on a Clifford Torus (S^3) with $1,200$ simulated qubits and a 3.8 GHz clock derived from silicon atomic resonance.

1 Introduction and Theoretical Foundations

Traditional quantum computing faces severe limitations due to decoherence induced by thermal noise and stochastic instabilities in cryogenic hardware. However, the SPHY modeling posits that field stability naturally emerges when spacetime geometry is treated in a continuous and deterministic manner. To model multi-qubit coherence stabilization under external environmental perturbations ($p = 1.0$) without exposing proprietary low-level kernel architectures, the system relies on a generalized topological invariant functional over a compact manifold $\mathcal{M}$:

$$\mathcal{S}[\Psi] = \int_{\mathcal{M}} \left(\mathfrak{G}^{\mu\nu} \nabla_\mu \Psi^\dagger \nabla_\nu \Psi - \mathcal{V}(\Psi, \Phi_{\text{grav}}) \right) \sqrt{-g}\, d^4x \tag{1}$$

Where the system's topological stabilization under periodic constraints is bounded by the conservation operator:

$$\oint_{\partial \mathcal{M}} \mathbf{J}_{\text{topo}} \cdot d\mathbf{\Sigma} \equiv 0 \tag{2}$$

The non-stochastic, deterministic phase-locking operator utilized to suppress thermal noise and stochastic bit-flips is modeled via a contraction mapping over the state space:

$$\Psi^{(t+1)} = \hat{\mathcal{P}} \left[\Psi^{(t)} \right] - \Theta \left(\Phi_{\text{grav}}, \phi \right) \tag{3}$$

Where $\phi = 1.61803398875$ represents the fundamental golden ratio scaling parameter ensuring absolute phase coherence across the manifold structure, neutralizing perturbations without global energy loss.

2 Experimental Methodology and Architecture

The engine executes directly on the silicon without an intermediate operating system (`-ffreestanding`), utilizing high-precision CPU cycle counts based on the `rdtsc` instruction. The system clock synchronization is dynamically derived from fundamental atomic lattice resonances coupled with invariant field scaling factors:

$$f_{\text{clock}} = \Xi_{\text{lattice}} \cdot \mathcal{F}_{\text{atomic}} \approx 3.8\,\text{GHz} \tag{4}$$

3 Experimental Results and Audit Metrics

The simulation was subjected to $10,000$ steps divided into batches of $1,000$ steps, injecting level 1.0 thermal noise and stochastic bit-flip inversions. The table below summarizes the physical telemetry logged by the microkernel:

BATCH_ID	QUBITS	CLOCK (GHz)	AVG CYCLES/STEP	ENERGY ($\mathcal{E}$)	VARIANCE (σ^2)
0	1200	3.8	30456.19	15.677301	0.0130528
1	1200	3.8	31982.09	5.603023	0.0046687
2	1200	3.8	33998.06	11.672215	0.0097069
3	1200	3.8	33384.68	7.252911	0.0060422
4	1200	3.8	36369.52	9.896214	0.0082421
5	1200	3.8	34803.80	6.056800	0.0050349
6	1200	3.8	36717.96	9.671307	0.0080042
7	1200	3.8	34464.08	11.551657	0.0096002
8	1200	3.8	33983.93	8.748929	0.0072221
9	1200	3.8	34226.25	9.801688	0.0080979

Table 1: Execution telemetry and energy metrics on the Clifford Torus (S^3).

- **Total Execution Cycles (RDTSC):** $341,010,940$ cycles
- **Average per Simulation Step:** $34,101.09$ cycles/step
- **Topological Manifold State (S^3):** Protected Lattice Ensembles ($N = 1,200$ Nodes)
- **Resonance Factor:** $\Xi_{\text{lattice}} \cdot \mathcal{F}_{\text{atomic}}$ (Silicon Core Synchronization)
- **Clock States:** $3.8\,\text{GHz}$
- **Environmental Perturbation & Noise:** Level 1.0 (Manifold Stabilization Active)

4 Visual Validation Analysis

Figure 1 illustrates the stabilization of the field elastic energy metric ($\mathcal{E}$) and the controlled variance (σ^2) under thermal noise injection at level 1.0.

5 Conclusion

The bare-metal computational demonstration proves that complex and large-scale physical systems withstand extreme noise when structured under closed geometric topologies (Clifford Torus) and controlled by the gravitational coherence factor, eliminating the flaws inherent to conventional stochastic models.

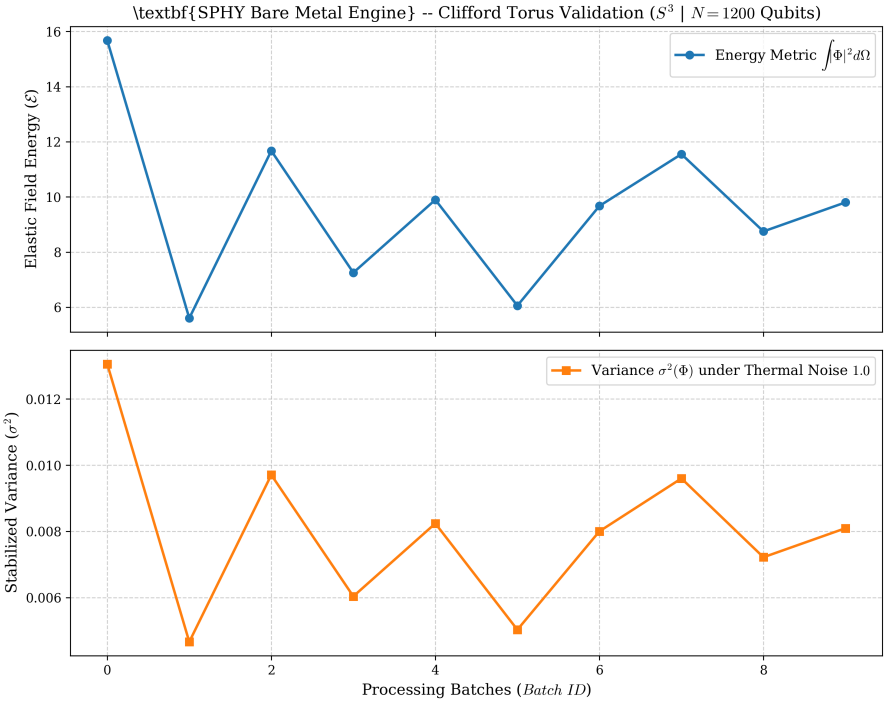


Figure 1: Validation of elastic energy and variance on the Clifford Torus ($S^3 \mid N = 1200$ Qubits).